

Element K - Reflection

Defining the Problem

Our task was to create a model home and develop a fire protection system capable of detecting fires, notifying occupants, and suppressing flames quickly while minimizing damage to life and property.

Ideation

Final Idea: We designed a fully automatic system using Arduino and Arduino software. This system features a two-factor detection system that includes an infrared sensor and an MQ2 gas sensor. Upon detecting a fire, the system activates a buzzer to alert residents and engages a suppression system that uses a pump to distribute water through piping installed throughout the house to extinguish the fire.

Selection and Prototype Design Requirements

Before construction of the model began, we created a cardboard prototype to determine the correct dimensions of the house. We also made cardboard prototypes of the furniture. These prototypes allowed us to finalize a design and served as a testing ground for our ideas, enabling us to see what worked without permanent consequences or wasted materials.

Prototype Building

The prototype took two weeks to build and involved collaboration among all team members. We used only cardboard for all components and produced several replicas to visualize the potential layout of the piping for the suppression system. Although we designed a platform for the Arduino board, it did not make it to the final build.

Testing

We conducted multiple tests of the detection system using controlled fires under adult supervision, varying conditions such as the presence of a fire source and smoke. We

successfully achieved our goal of activating the detection and notification system within 30 seconds. However, we faced challenges testing the suppression system. We managed a few tests that confirmed the functionality of the pump and piping, but we were unable to test the entire system together due to time constraints. This led to some errors, but we ultimately resolved them.

Evaluation

Our system was effective in quickly detecting and notifying users about a fire, and it was able to extinguish the fire in a reasonable timeframe. However, during our demonstration, the suppression system did not activate automatically; it required manual intervention to reconnect two wires for proper operation.

By the end of this process, I learned that making mistakes and iterating is a natural part of the design process. We should use feedback to continually improve our work.